

Introduction to Kubernetes

Official definition of Kubernetes



- ▶ Open source container **orchestration tool**
- ▶ Developed by **Google**
- ▶ Helps manage containerized applications



What features do orchestration tools offer?



- ▶ High **Availability** or no downtime



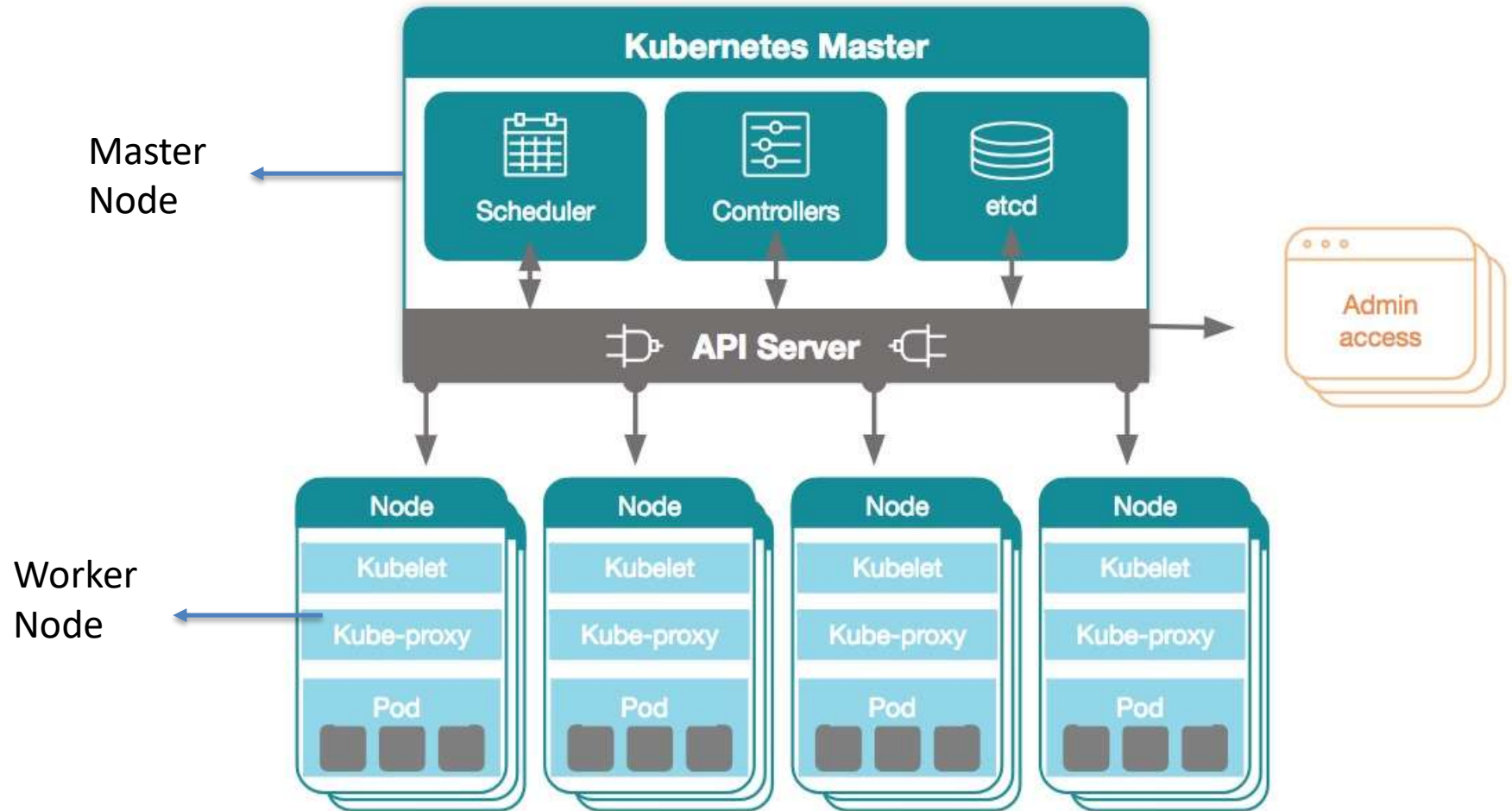
- ▶ **Scalability** or high performance



- ▶ **Disaster recovery** - backup and restore



Kubernetes Architecture



Core Components

Control Plane Components(Master Node)

Manage the overall state of the cluster:

- ✓ [kube-apiserver](#)

The core component server that exposes the Kubernetes HTTP API

- ✓ [etcd](#)

Consistent and highly-available key value store for all API server data

- ✓ [kube-scheduler](#)

Looks for Pods not yet bound to a node, and assigns each Pod to a suitable node.

- ✓ [kube-controller-manager](#)

Runs [controllers](#) to implement Kubernetes API behavior.

- ✓ [cloud-controller-manager](#) (optional)

Integrates with underlying cloud provider(s)

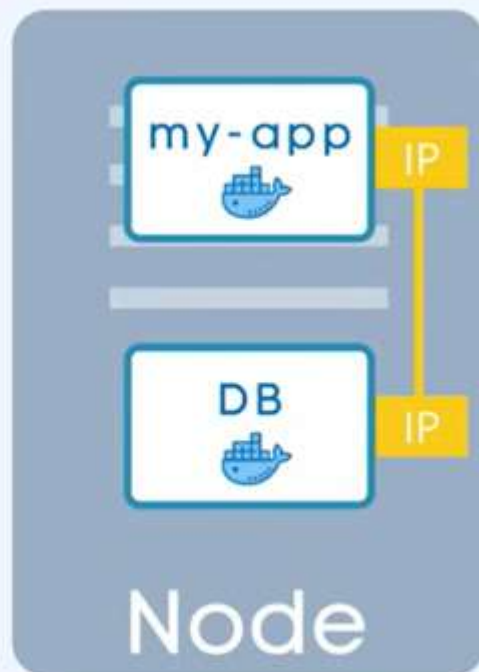
Core Components

Node Components

Run on every node, maintaining running pods and providing the Kubernetes runtime environment:

- ✓ **kubelet**
Ensures that Pods are running, including their containers.
- ✓ **kube-proxy (optional)**
Maintains network rules on nodes to implement Services.
- ✓ **Container runtime**
Software responsible for running containers.

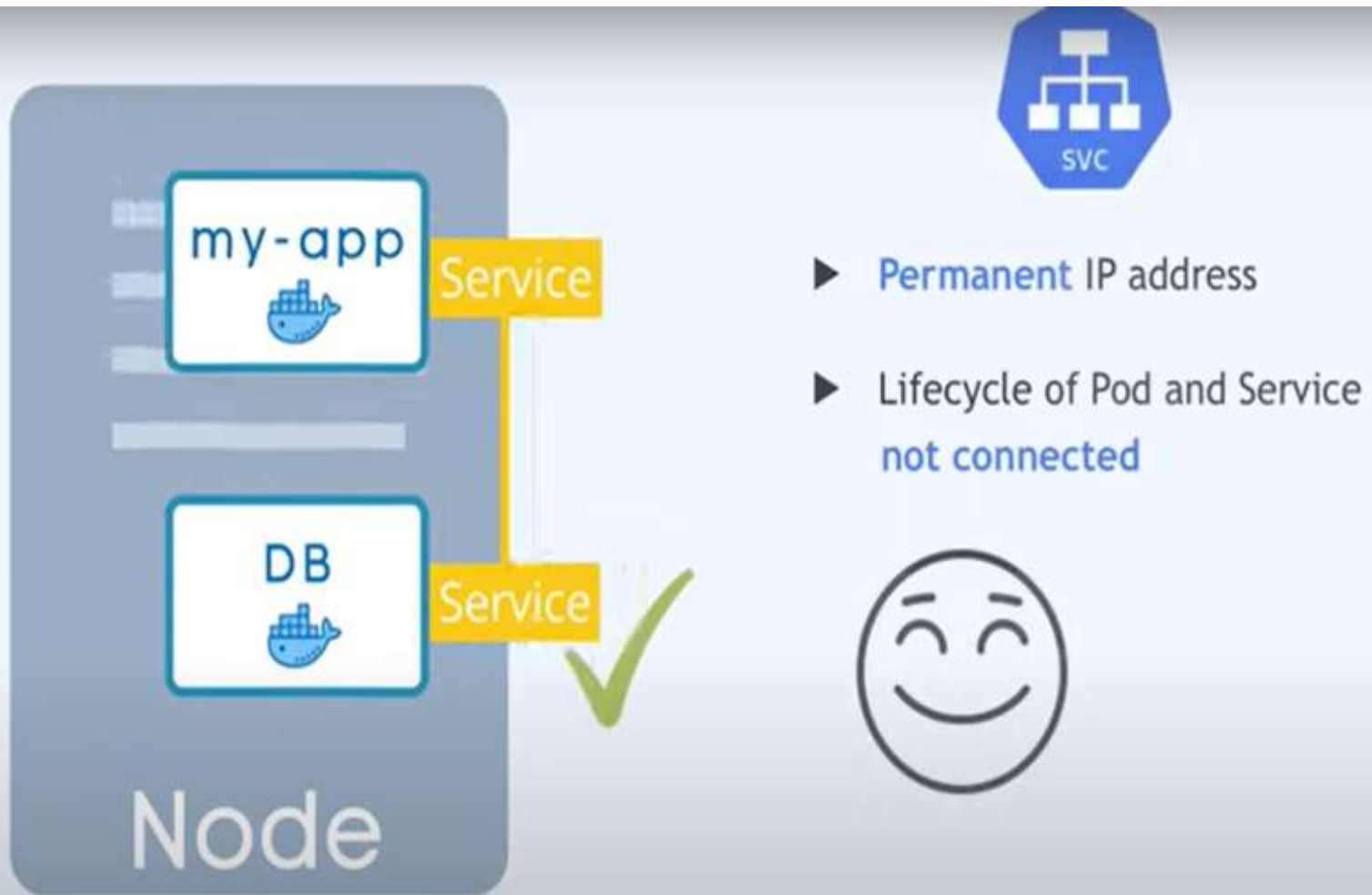
Kubernetes Pods



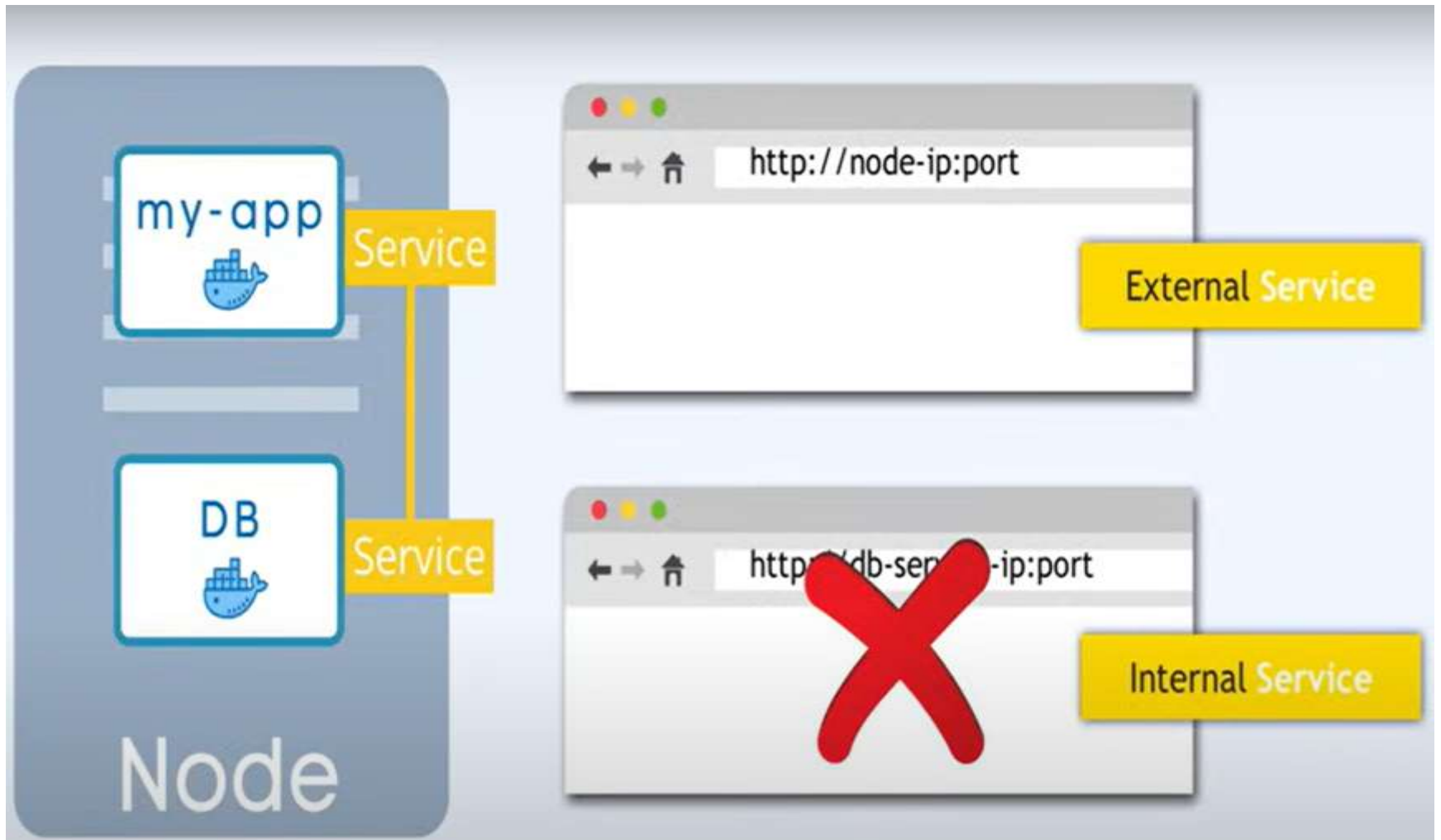
- ▶ **Smallest** unit in Kubernetes
- ▶ **Abstraction** over container
- ▶ Usually **1 Application** per Pod
- ▶ Each Pod gets its **own IP address**

Pods are **ephemeral!**

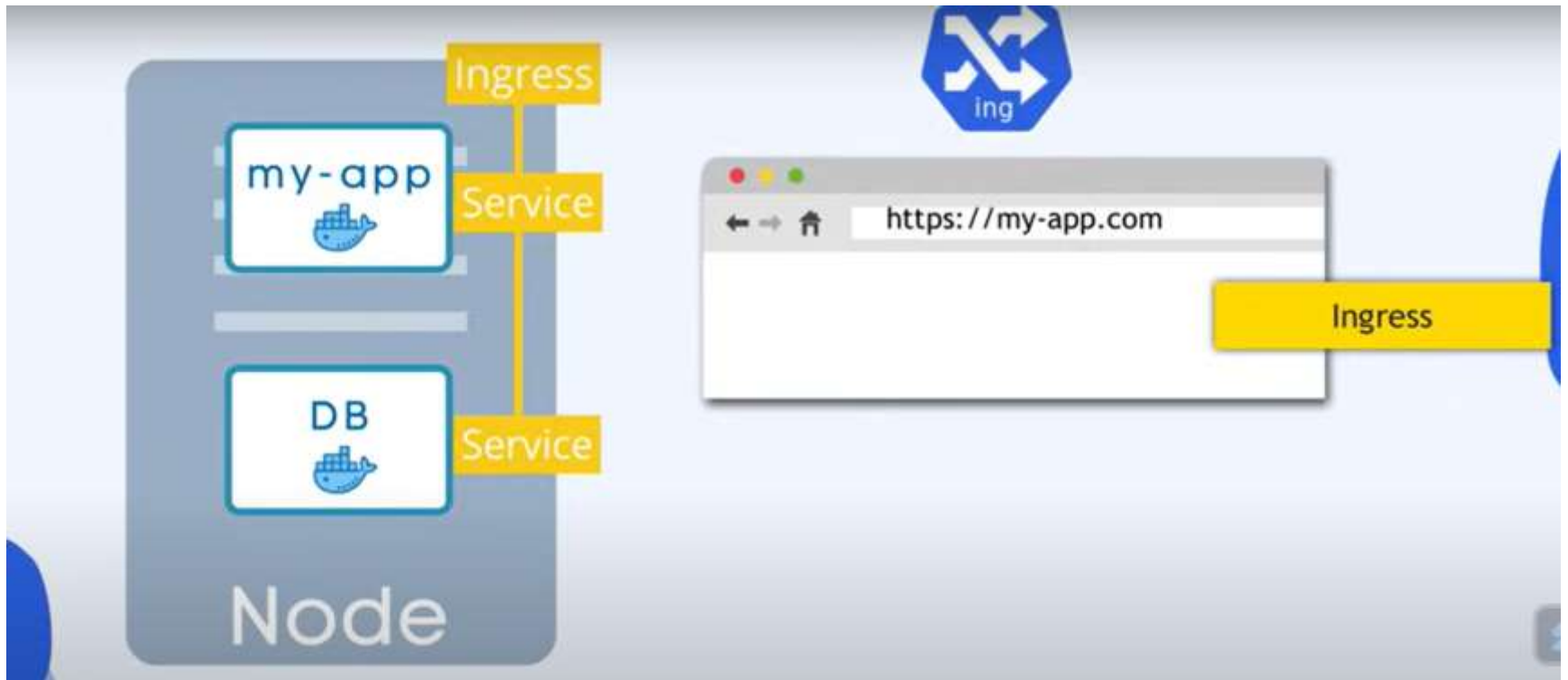
Service



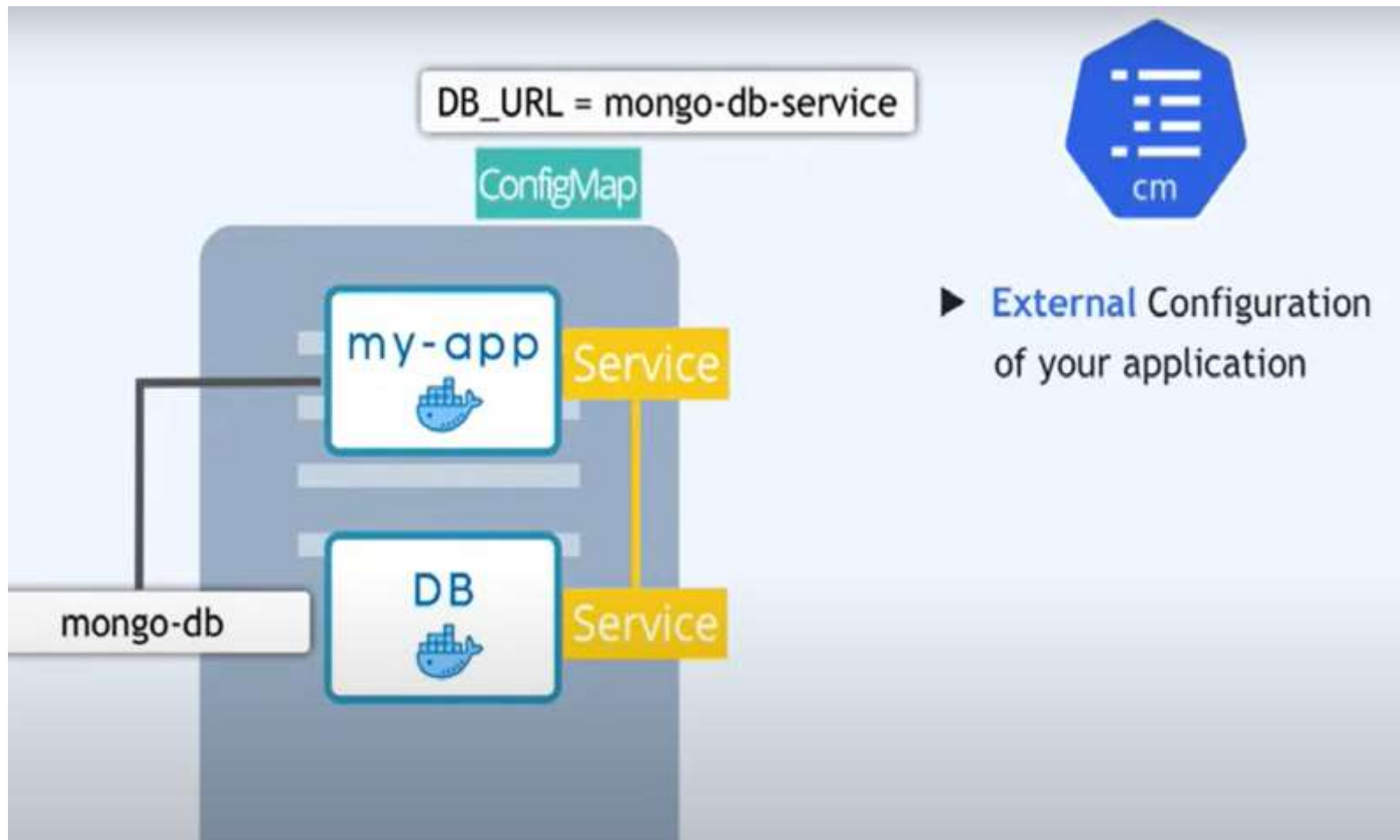
Service



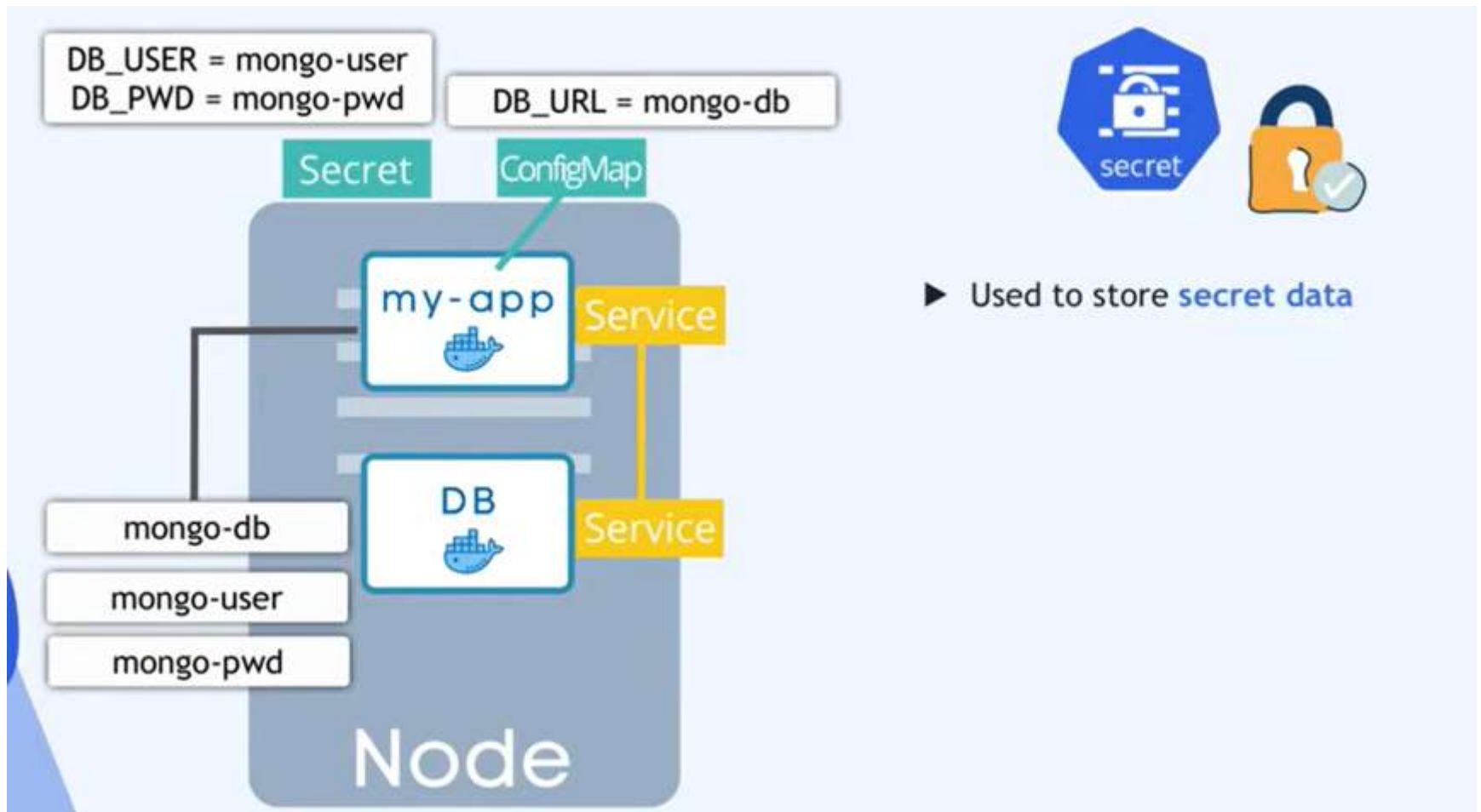
Ingress



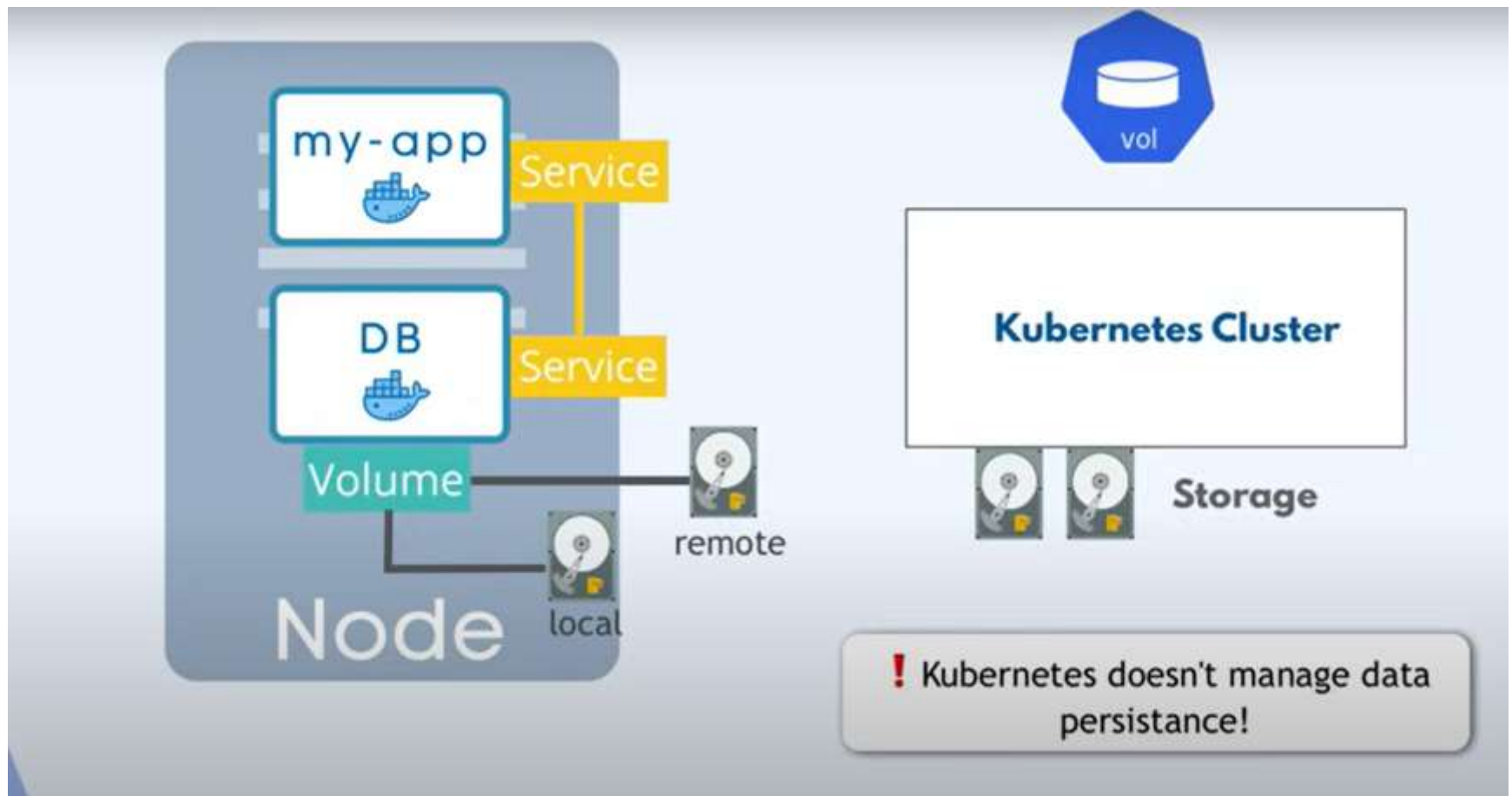
Configmap



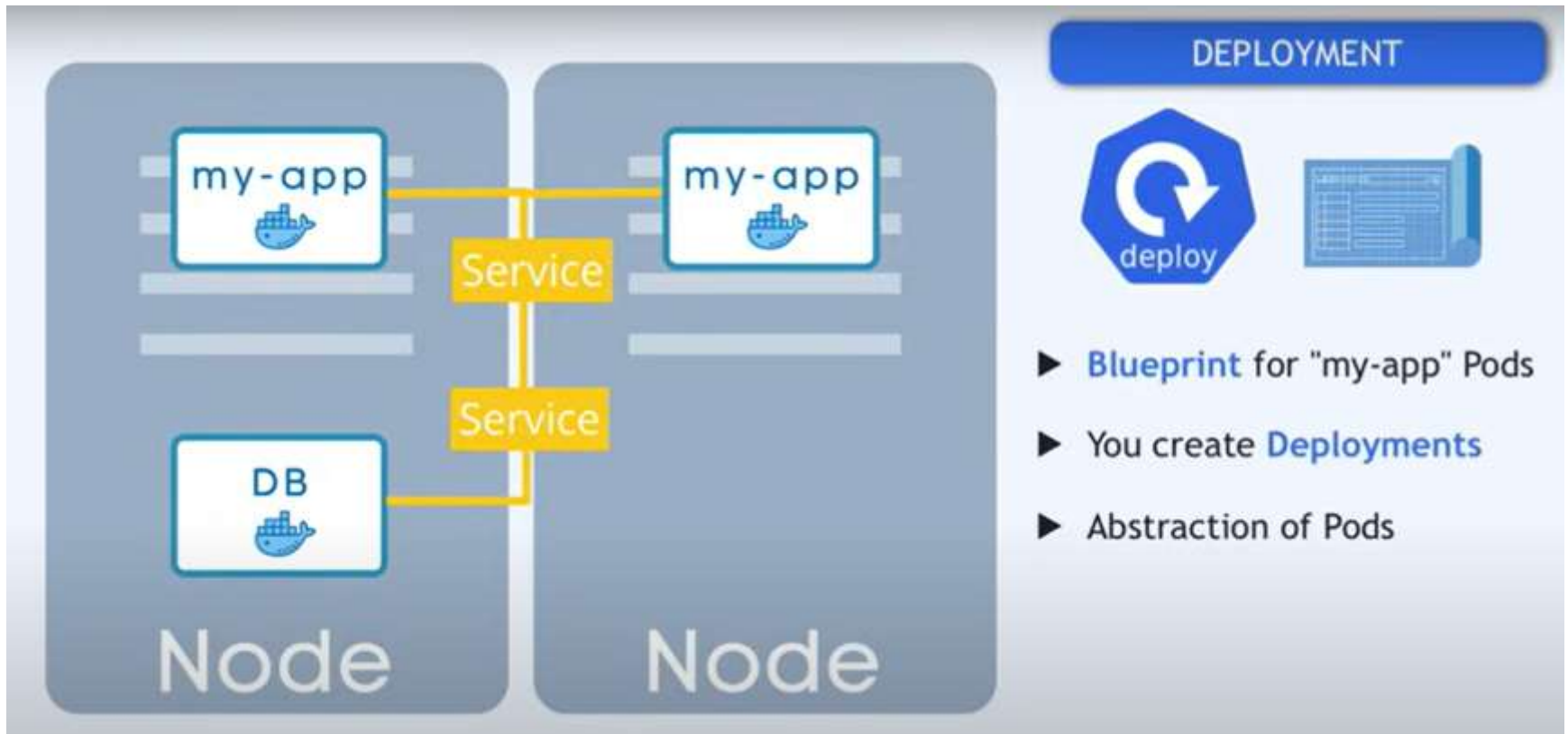
Secret



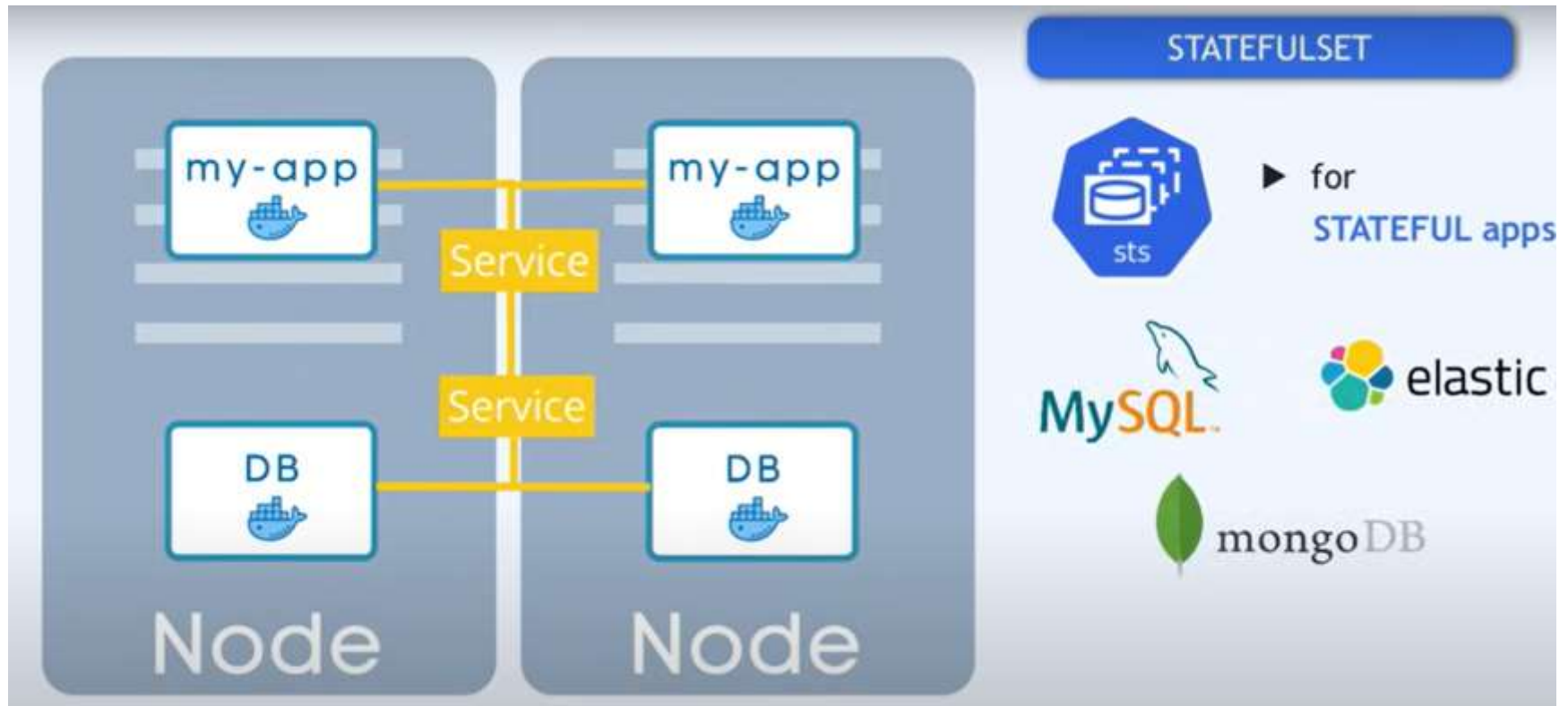
Volumes



Deployment



STATEFULSET



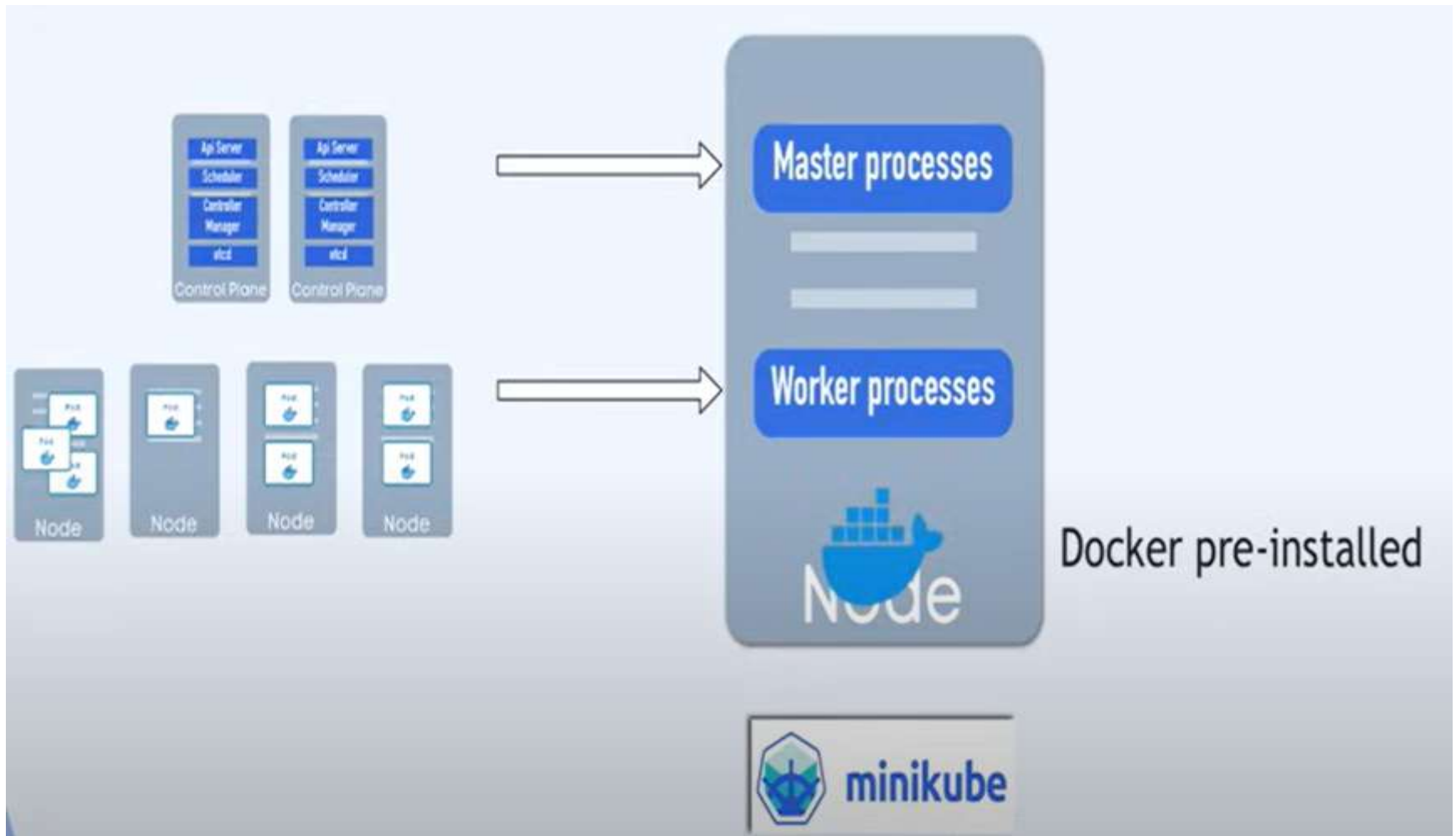
Summary

- ✓ Node -> Master Node , Worker Node
- ✓ PODs, Service, Ingress
- ✓ Configmap, Secret
- ✓ Volume
- ✓ Deployment, Statefulset

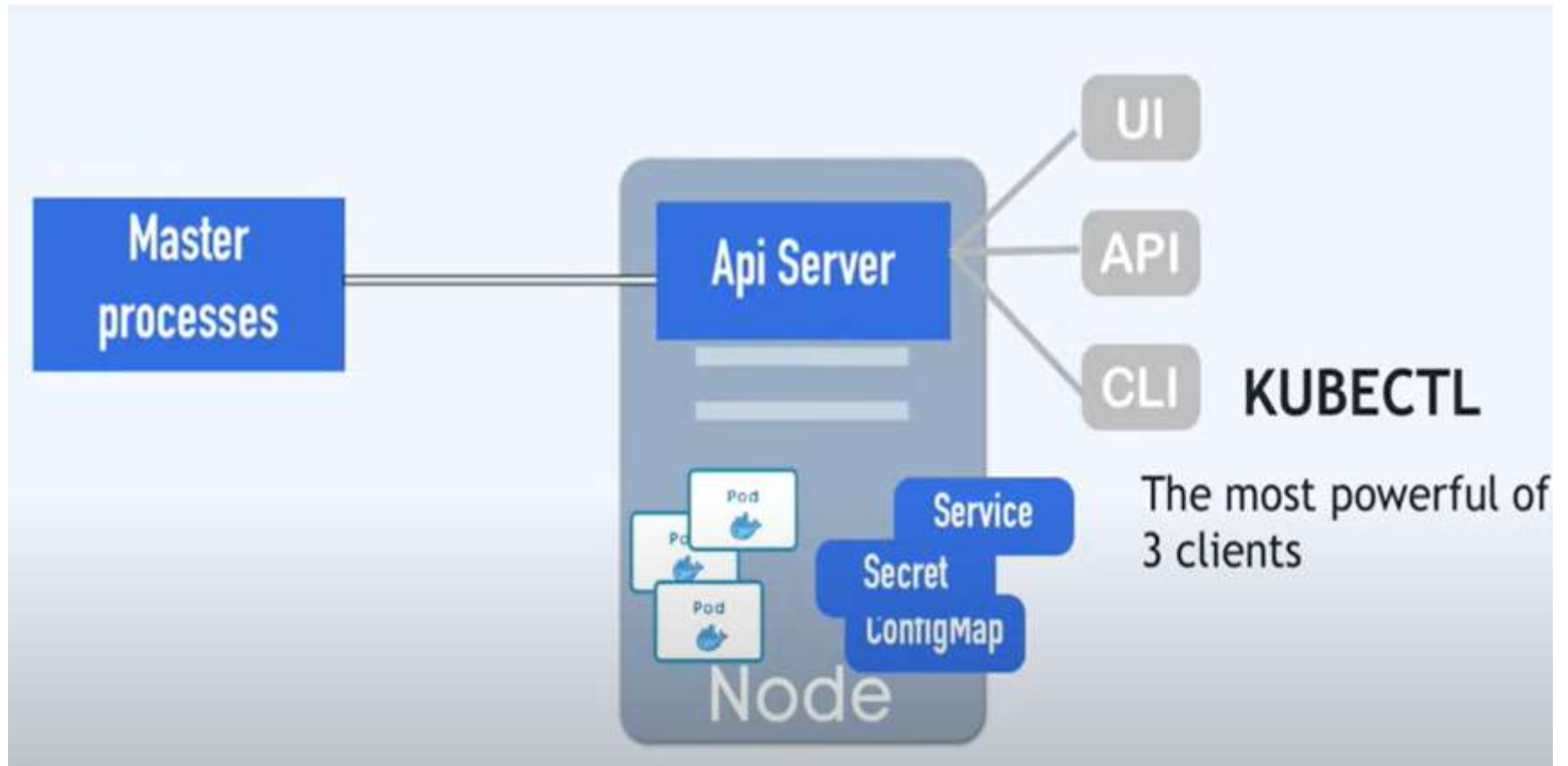
Demo Setup

- Minikube: Open-source tool, one node setup for running K8s cluster (Master Node and Worker Node on single Node)
- Kubectl: Command Line tool to communicate to KubeAPI

Minikube



Kubectl



Demo

